On the Rival Nature of Data: Tech and Policy Implications

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Abstract

Data is often thought of and treated as a non-rival good, which would imply that one person's use of data does not inherently diminish its availability for others. Building on research in privacy and statistics, we argue that there exist many important settings in which data should be treated as a rival good. Our argument takes into account modern uses of data for statistics, machine learning, and a variety of other purposes, in conjunction with requirements of privacy protection and statistical validity. Excessive sharing or reuse of data about individuals can lead to leakage of sensitive personal information, potentially causing harm to those whose information is included in the data. Overuse of data in statistics or machine learning can lead to overfitting, i.e., models that perform well on training data but poorly on fresh unseen data.

Recognizing the rival nature of data offers an opportunity to rethink the way data are managed and used. In an age where the training of AI models generates a massive appetite for data, this perspective has the potential to inform the creation of new regulation and technical infrastructure that will be able to safely and responsibly manage data, track their various uses, and ensure that privacy and statistical usefulness are respected and preserved.

We observe that current EU regulation and existing approaches to science seeking to increase opportunities for data-sharing and reuse of data misconstrue the complex nature of data, inadvertently creating risks of privacy harms and overfitting, hence squandering the societal benefits that can be derived from data.

Recognizing the rival nature of data has implications for policy and practice. Regulation should address the limitations and risks associated with data reuse and facilitate technological measures to track, analyze, and manage data usage with the goal of ensuring that privacy and statistical validity are maintained.

Keywords

data, privacy, statistical validity, Data Governance Act (DGA), privacy budget

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1 Introduction

Data have become a cornerstone of the modern economy, driving innovation, informing decision making, and reshaping societal interactions. As with other economic goods, understanding the nature of data is a crucial first stage in designing an appropriate public policy approach towards their management. The nature of a good influences the way it can be generated, allocated, used, and governed. Data have long been thought of as a non-rival good, whose use by one party does not restrict its availability for use by others. In this Article we argue that there are common settings where concerns of privacy and statistical validity require regulation to treat data as a rival good. We hence establish regulatory principles that promote data use as a rival good in these contexts.

This insight has implications. Recent developments, particularly in the field of AI, have once again drawn attention to the role of data as a driving force for modern developments and innovative technologies. These have highlighted that even the gargantuan amounts of existing data are finite, and that generating new data can be a complex and expensive task. Companies need more and more data to train their AI models, and may increasingly attempt to exclude other companies from accessing the same training data, including using various legal mechanisms. The rival nature of data, coupled with an increasingly growing appetite for data, requires us to think about how data should be regulated and managed, and how their use should be tracked through different settings.

1.1 Rival and Non-Rival Goods

The concept of rival goods was first laid out by Vincent Ostrom and (Nobel prize winner) Elinor Ostrom in a book chapter titled "Public Goods and Public Choices" [37]. **Rival goods** are those that can be consumed by no more than one person at a time. The use of a rival good by one person significantly reduces the ability of other parties to enjoy that good. **Non-rival goods** are those that can be used by more than one person simultaneously. The use of a non-rival good by one person does not impact the ability of others to simultaneously enjoy the non-rival good. In many cases, the cost of providing a marginal consumer with access to a non-rival good is relatively low, while the cost of providing a marginal consumer with access to a rival good is more similar to the cost incurred in providing the good to the first user.

Examples of rival goods include an apple (if Alice eats it, Bob cannot), a seat on a particular airplane flight (when Bob sits it in, Alice cannot), and an Internet domain name (if Carol pays to register a certain domain name, Bob has to wait for her license to expire before he can register the same domain). Examples of non-rival goods include clean air and national security (which are also public

goods), radio or TV broadcast, and digital music (millions can listen to the same Taylor Swift song at the same time).

The rival nature of goods is of course not binary and instead lies along a spectrum. For example, while highways can largely be characterized as a non-rival good, extreme overuse of a highway can have a strong impact on the ability of a marginal user to enjoy the highway. Public parks and fireworks shows have similar characteristics: while first users do not interfere with each other's ability to enjoy these goods, overcrowding will reduce the marginal user's ability to access and enjoy the park or view fireworks. In such cases, the law plays a meaningful role in allocating access to the good; to manage such challenges, governments may require entrance fees to enter a park or build toll booths along a highway.

Characterizing a good as rival impacts the policy considerations around it. Since the consumption of a rival good by one person precludes its simultaneous consumption by another, such goods are more likely to be characterized by scarcity than non-rival goods. Thus, free markets are more likely to facilitate the provision and allocation of rival goods. Property rules and licensing regimes assist and encourage such behavior.

1.2 Excludable and Non-Excludable Goods

The concept of excludability is used to categorize goods and services based on the ability to restrict their consumption. **Excludable goods** are those that can be withheld from some consumers, e.g., non-payers. A movie theater, for example, can prevent those without a ticket from entering and enjoying the film and a farmer can fence off their orchard to exclude people from taking apples without paying. That is, there are mechanisms in place that make it possible to exclude some consumers.

Non-excludable goods are those where it is difficult or impossible to prevent consumption. A public fireworks display is an example: Once the fireworks are launched, anyone nearby can enjoy the show. Similarly, national defense provides protection to everyone within a country's borders (and regardless of their tax contributions).

Excludability can induce a rival nature for some goods, and non-rival goods are also often non-excludable. However, the two concepts are different. For example, it could be infeasible to put a fence around a lake to prevent fishing in it (and hence fishing there is non-excludable), but there could still be a limited number of fish in the lake (making it rival). Excludable goods are also not always rival: cable television uses technology to allow only subscribers to view their channels, but one person's viewing a cable channel does not negatively impact another person's ability to enjoy it.

1.3 Characteristics of Data

Data consist of collections of digitally encoded observations about phenomena of interest. Data serve as the foundation for reporting, research, and decision-making by government agencies, businesses, researchers, and the public at large. When analyzed with care to ensure that findings from a particular dataset generalize from the collected data to the larger population, data analysis provides valuable evidence-based conclusions, trend identification, optimization, outcome prediction, and more. In an increasingly digital world, data have become a crucial resource in both the public and private sectors, shaping strategies, driving progress, and underlying service improvements, making data increasingly more valuable. Some even argue that data have replaced oil as the world's most valuable resource.[1]

It can feel natural to conceptualize data as non-rival, in part because intuition suggests that human-derived data are plentiful (for example, more than 500 hours of content are are uploaded to YouTube per minute)[2] and that facilitating access to data implies making better use of it in ways that benefit everyone. Intuitively, it seems reasonable that data ought to be more similar to digital music than to an apple – once data is collected, it can be analyzed without limit.

As data occupies growing importance in the market, economic research has become increasingly curious about the nature and value of data [43]. In search of useful models and analogies, economists have compared data to labor [10, 39], to pollution [11], to various types of infrastructure [35], and to public goods [29].

A review of the scholarship analyzing the nature of data reveals that several scholars have characterized data as a non-rival good. Varian characterizes data as a non-rival good, noting that this nature of data has practical advantages. They argue that an algorithm trained on data from a single hospital is likely to be much less accurate than one trained on medical data from all hospitals in the US, while (purportedly) not detracting from the ability of other parties to use the same data for their own purposes [42]. Jones and Tonetti note several characteristics of data that also lead them to classify them as non-rival. They argue that unlike a kilogram of rice, or an hour of an accountant's workday, which diminish when used, "data can be used by any number of firms or people simultaneously, without being diminished" [30]. They argue that the ability of many parties to make use of the same data can have positive societal impacts. Recently, Veldkamp and Chung joined the group of researchers who characterize data as non-rival good. Analogizing data and technology, they observe that a single idea or technology can be used by many firms simultaneously [43]. The classification of data as a non-rival good leads Varian to the conclusion that discussions regarding the ownership of data should be replaced by discussions regarding access to data.

1.4 This Article

We agree that when there is no reason to restrict access to data, it is non-rival. However, in Section 2 we review technical results from the last two decades and interpret them as showing that in many typical settings, access to data should be limited in a way that makes them a rival good. Treating data otherwise leads to the loss of important social values: the privacy of individual information and/or the usefulness of data in providing reliable scientific findings and insights. We then, in Section 3, review the approach to data taken by the law and, in particular, we focus on the EU Data Governance Act (DGA) [4]. We observe that the DGA misconstrues

[1] *The World's Most Valuable Resource is No Longer Oil, But Data,* Economist (May 6, 2017), https://www.economist.com/leaders/2017/05/06/the-worlds-most-valuable-resource-is-no-longer-oil-but-data.

[2] Viraj Mahajan, *100+ YouTube Statistics in 2024: Users, Revnue & More*, Notta (Mar. 19, 2024), https://www.notta.ai/en/blog/youtube-statistics.

the nature of data as it fosters a data ecosystem where a large number of entities can collect and share data without the coordination needed to control and limit its use.

We argue that regulators need to adopt a new approach to data, where privacy and statistical validity are treated as finite consumable goods (and hence, rival). In Section 4 we review directions for regulatory changes and technical challenges towards creating an infrastructure where information about data sharing is used to coordinate data use such that privacy and/or statistical validity are preserved and to prioritize uses, e.g., via the deployment of data markets.

2 On the Rival Nature of Data

We discuss findings from two literatures that have established formal mathematical frameworks for reasoning about key desiderata for data. The first, on privacy, has established mathematical limits on how data can be used if we wish to prevent the underlying data from being revealed. The second, on statistical validity, has established a mathematical toolkit for reasoning about the ways in which a series of statistical analyses of a dataset can lead to "discoveries" that are actually quirks of the dataset that would not replicate on a fresh dataset (we call making such "discoveries" *overfitting*).

In both cases, mathematical models reveal a need to treat privacy and statistical validity as finite consumable resources and associate them with a "budget" that is depleted with every use of a dataset. This has consequences for the responsible use and management of data and creates a need for making the use of data rival.

2.1 How Privacy Concerns Make Data Rival

Privacy is a fundamental social value that enjoys legal protection in many jurisdictions. An individual who enjoys strong privacy protections is free to communicate and express herself, explore new thoughts and ideas, and make decisions based on her values and desires and not based on fear of facing consequences. In the United States, the right to privacy comes from the fourth amendment of the Constitution. In the EU, privacy is recognized as a fundamental right in Article 7 of the European Charter of Fundamental Rights, as well as in Article 8 of the European Convention of Human Rights.

Protecting privacy as a fundamental right supports a key principle of human dignity, upholds an essential part of individual autonomy, and allows people to make decisions about their lives in a way that respects their liberty [27]. The detailed information collected about individuals' online activity, communication patters, location, medical needs, financial status, and much more is an invaluable source for learning about human behavior, health, decision making, etc. and for providing individuals with services adjusted to their particular needs. It has also become a major privacy concern and modern regulations put forward privacy and data protection requirements for organizations. The actual protection of individuals' data privacy is predominantly realized by technological safeguards in the form of privacy-enhancing technologies (PETs).

2.1.1 A Toy Example – Differencing Attack. Assume that a research institution collects information about Washington, DC residents with the purpose of making it available to demographic researchers. Suppose that Alice and Bob are independent data analysts interested in counting the number of DC residents who who are HIV+. Imagine that Alice issues queries the dataset, and in particular asks for the exact count c_A of people who are HIV+. Alice plans to submit her finding for publication in a conference and she is aware of the sensitive nature of the data. For the count c_A, she reasons that it would not allow the identification of individuals, as the city population is quite large and being HIV+ is not very rare. Independently, Bob uses similar reasoning and queries the data for the number of residents who are HIV+, but Bob asks the count not to include a specific individual, Charlie, as Charlie is a well-known public figure, and Bob practices extra caution to protect the privacy of Charlie's health information.

While each of the reasonings by Alice and Bob makes sense in isolation a, observe that if both counts c_A, c_B are published it, then the difference $c_A - c_B$ equals 1 exactly if Charlie is HIV+ and 0 otherwise. This mode of disclosure is referred to in the privacy literature as a *differencing attack*.

Stated differently, although each of the exact answers c_A, c_B likely preserves Charlie's privacy in isolation, the *composition* of exact answers to Alice's and Bob's queries results in a privacy breach. This toy example establishes that, at least in some circumstances, access to data should be restricted, and hence rival. To preserve privacy, if Alice's query has been allowed, Bob's query should be denied.

The existence of privacy and data protection laws and regulations in the US and the EU suggests that legislators intend to provide individuals with protection of their sensitive data, and in particular from the possibility that their sensitive information would be leaked. As a few examples, in the US personal medical information is protected by the Health Insurance Portability and Accountability Act (HIPPA) privacy rule, and Children's online privacy is protected through the Children's Online Privacy Protection Act (COPPA). In the EU the General Data Privacy Regulation (GDPR) protects personal data at large. However, it is hard to tell whether adhering to these regulations would prevent the above-mentioned differencing attack, especially considering that Alice and Bob as well as other uncoordinated parties may process information collected about the same or similar populations.

2.1.2 The Fundamental Law of Information Recovery. The problem described above can be countered by introducing random perturbation in the responses to Alice's and Bob's queries, so that the released (perturbed) counts do not make it certain whether the excluded person p has the trait or not. However, there are settings where the privacy problem is inherent. In particular, there are settings where *reconstruction attacks* become inevitable. These are situations where a party with access to data releases made based on a dataset (or several related datasets) can recover the sensitive values of one or more individuals with good accuracy and high certainty.[3]

As a concrete example, consider analysts making statistical queries to the Washington, DC resident dataset. An analyst may specify a subset of the city residents, e.g., by combining some non-private characteristics (age group, sex, neighborhood, etc.) and ask for the

[3]The formal mathematical definition of reconstruction attacks is nuanced and context dependent. *See* [15] for a definition of reconstruction attacks and a discussion of the subtleties in formalizing the concept.

(approximate) number of those in the subset who have the sensitive trait. Such queries are the fundamental bread and butter of statistical analysis and machine learning.

Dinur and Nissim proved that even if answered with significant noise, answering many such statistical queries inevitably leads to accurate recovery of the underlying sensitive data [19], a result which was further strengthened in follow-up work [22, 24].[4] This is not merely a theoretical result – such an attack has been successfully applied to retrieve sensitive data from a practical commercial system that was designed to satisfy the GDPR's anonymization requirements [14]. These and other similar findings led Dwork and Roth to articulate the *Fundamental Law of Information Recovery*:

> "[O]verly accurate answers to too many questions will destroy privacy in a spectacular way" [23].

Importantly, the fundamental law of information recovery applies regardless of any techniques that might be deployed to improve the privacy of the analysis. In other words, regardless of whether or what technique is used to ensure privacy, the *composition* (combination) of informative responses to a series of analyst(s) queries will eventually result in a breach of privacy. Every piece of useful information gleaned from the data contributes to a cumulative erosion of privacy, which can eventually suffice to infer personal information with high certainty.[5]

Reconstruction attacks hence establish that there exist natural settings, such as as that of making statistical queries to a dataset, where privacy cannot be maintained if all queries of interest can are answered, even if not with perfect accuracy, i.e., settings where imposing privacy requirements make data inherently rival. In such settings, there are fundamental, unavoidable tensions and tradeoffs between answering some queries and answering others, or between the accuracies with which each query can be answered.

2.1.3 Privacy and the Need to Restrict Data Use. Both theory and practice highlight that recent rapid growth in the collection and use of sensitive personal data magnifies the risks of taking a short-sighted approach to personal privacy and surveillance that ignores the effects of composition. However, analyzing how the combination of various data uses erodes privacy (and hence making decisions based on such analysis) proves to be extremely difficult. In particular, traditional data protection frameworks are not equipped with tools for making such determinations.

As one example, the HIPAA safe harbor privacy rule specifies the removal of 18 types of identifiers as sufficient (if "No actual knowledge residual information can identify individual").[6] It is difficult (and likely impossible) to quantify the effect of a single data release under HIPAA's safe harbor rule, not to speak of the cumulative effects of multiple such releases. Another example is the framework of k-anonymity, under which data releases contained constrained (or redacted) data so that the potentially identifying information of each individual in the release is identical to (and hence indistinguishable from) the information of at least $k-1$ other individuals in the release [40]. Setting $k = 1$ provides no guarantee of anonymity, and as k grows, anonymity seems to improve. However, although the parameter k quantifies a property of the release, it is insufficient for quantifying the effect one or more k-anonymized data releases have on individual privacy. Furthermore, it has been shown in theory and in practice that the composition of multiple k-anonymized releases can result in a total collapse of k-anonymity (i.e., effectively bringing k down to 1) [13, 26].

Differential privacy [21][7] is a formal mathematical framework equipped with a parameter bounding privacy loss. This parameter limits the potential privacy loss to each individual due to the inclusion of their data in a differentially private analysis. In fact, with differential privacy, it is often possible to reason about tradeoffs between the accuracy with which an analysis can be performed and its privacy loss (the better the privacy, the less accurate the analysis, and vice versa.) Furthermore, differential privacy is currently the only framework that provides a calculus that bounds the loss of individual privacy resulting from the composition of several analyses. This ability to reason about how privacy losses accumulate across data uses means that a *privacy budget* may be set for one or more datasets containing information collected from a subject; the budget should express an acceptable level of privacy loss. The privacy budget may then be divided between analysts who will perform computations (with differential privacy guarantees) over the relevant datasets. This budget, shared between multiple parties, makes the rival nature of the data explicit.

We emphasize that the Fundamental Law of Information Recovery of Dwork and Roth holds regardless of the disclosure limitation techniques in use. The benefit of differential privacy is that it is possible to reason about accuracy vs. privacy loss tradeoffs rigorously, and in a quantified manner.

2.1.4 Example: the US 2020 Decennial Census. The rival nature of data raised by privacy concerns is nicely illustrated by recent events at the US Census. Using data from statistical tables published for the 2010 decennial US Census in the US, researchers reconstructed the data respondents provided to the US Census Bureau with significant precision—an exact recovery of sex, age, race, ethnicity, and block-level geography for 46% of the US population, and for 71% of the population, their sex, race, ethnicity and geography were recovered precisely, and their age was recovered within a one year. Furthermore, by linking the reconstructed information with data that were commercially available at the time of the 2010 Census, the researchers could accurately re-identify the responses of 52 million respondents [8]. More recent work provides attacks that also rank the reconstructed respondents' records by their confidence of appearing in the underlying Census data, thus revealing which individuals' data are precisely reconstructed with high confidence [18].

These findings led the US Census to redesign the releases for the 2020 Census within the framework of differential privacy. This forced the Census to establish a privacy budget (deciding how much privacy would be guaranteed to individuals whose data were gathered) and therefore face explicit tradeoffs between which data products could be published, and with what accuracy.

[4] If the dataset contains information of n residents, and the query language is sufficiently rich, then $O(n)$ queries suffice for reconstructing the data with high accuracy, if a high enough portion of the queries are answered with error $\ll \sqrt{n}$ [19, 22, 24].

[5] *See* [25] for a more in-depth discussion of composition effects.

[6] Health Insurance Portability and Accountability Act § 164.514(b)(2). *See* https://www.hhs.gov/hipaa/for-professionals/special-topics/de-identification/index.html

[7] *See* [45] for a non-technical exposition of differential privacy.

Data published by the US Census form a crucial part of the US democracy data infrastructure. Census data are used in redistricting, determining congressional representation, allocating federal funds, making decisions by government agencies and private parties, and in research. The move to a differentially private release therefore raised concerns whether the released data would be suitable for use for these tasks, and battles have raged about whose analyses should be prioritized. In a court case highlighting the tension between ensuring privacy and maintaining data accuracy, the State of Alabama challenged the US Census Bureau's use of a differentially private data release for the 2020 Decennial Census, arguing that the techniques used by the Census compromised the accuracy of the data, potentially affecting redistricting efforts and the distribution of federal funds [7].

2.2 How Statistical Validity Concerns Can Make Data Rival

The ability to adaptively interact with a dataset is key to how we use data, how modern machine learning works, and how science progresses. By *adaptive*, we mean that the computations performed on a dataset are chosen based in part on the results of previous computations on those same data. Examples might include noticing that people in a particular geographic region seem to have a startling rate of a rare disease and then trying to explore whether that pattern is statistically significant (as the pattern that is analyzed is chosen based on having noticed people in the data with the disease, this analysis would not have been done without first observing that many people are sick); using a rich dataset of animal behavior to try to predict future migration patterns by first isolating the most relevant variables and then building a model based on them (the migration-prediction model would not have been based on these specific variables had they not shown relevance in the data); or gaining insight from another research group's use of a shared astronomy database and then using that insight to develop a new study on the same data (that exact study would not have been designed if not for the prior insights).

Adaptivity is key to making efficient progress in understanding the world. If researchers could not use prior insights to guide the questions they ask, researchers would be stuck having to explore *all possible* questions in order to discover the interesting ones and their results. However, at the same time, adaptivity comes with risks that threaten the validity of the scientific conclusions it enables.

Returning to the example of noticing a cluster of disease in a particular geographic region, suppose that a doctor noticed that a surprising number of people in the New Brunswick, Canada region were showing signs of neurological decline.[8] How can the doctor know whether this cluster of patients have a common cause, which ought to be investigated to save others from their fate, or whether this is simply a coincidence? On the one hand, dozens more patients than one would normally expect to see, with frightening symptoms, some of them leading to death, seems like cause for panic. On the other hand, if one considers all of the possible geographic regions in the world and all of the possible "rare" diseases one might note in them, there are so many possible combinations that it would actually be surprising if there *weren't* seeming clusters of *some* rare diseases in *some* regions, just by random chance. So what is the doctor to do? The choice to analyze the presence of neurological decline in New Brunswick, Canada was made as a function of informal observations drawn on the same data, and so it is easy to be misled—even if the number of observed cases is indeed far outside the norm, it becomes difficult to distinguish whether this is due to random chance or a true underlying phenomenon.

This sort of quandary, where adaptivity can lead to invalid, uninterpretable, or misleading statistical conclusions, is actually a widespread problem, and it is blamed in part for a massive non-reproducibility crisis in many data-driven sciences [28]—in recent years, we have seen that many prominent papers in many fields draw conclusions that do not replicate when other groups of scientists attempt to do the same study but on a fresh group of subjects [12, 16].

This risk of undermining statistical validity through repeated, adaptive, use of data is a concern that manifests regardless of whether the objects of study are human (note that one could tell a similar story about one-eyed fish or about oddly shaped rocks). The concern is less a concern that the occasional scientific study may draw an erroneous conclusion—that is a phenomenon that can often be addressed by attempts to reproduce results—but rather a concern that a lack of infrastructure to promote the responsible use and reuse of datasets can result in squandering their potential to advance societal good. A cloud of statistically questionable results can pollute an entire research area, leading scientists to waste time and resources exploring unfruitful directions. Without protections in place, even scientists who are aware of the risks of adaptivity cannot know whether their own results or others' results are statistically valid, and hence cannot know whether to rely and build on them, slowing scientific progress.

2.2.1 Managing Adaptivity. What are scientists to do? One option widely embraced by certain scientific communities is to abandon adaptivity: Researchers are expected or required to *preregister* the analyses they plan to do, *before* they gather data, and then only the results of these preregistered analyzes can be published (or if there are other analyses, adaptively chosen, they must be marked as such) [36]. However, abandoning adaptivity also eliminates its benefits. Ideally, the reuse of valuable datasets that may have been difficult or expensive to gather should be allowed and promoted.

Another possible approach that has emerged from the theoretical computer science literature over the past decade is the possibility to develop new statistical techniques that prevent computations over a dataset from leaking too much about the specific data points in it, so future computations are limited in how much they can be misled by the prior results, but can still benefit from prior aggregate insights derived from the data [20, 31]. This literature, while still in its early days, hints towards a future where it might be possible to reuse data adaptively and safely, with protections and monitoring infrastructure in place, to help science make the most of its data.

2.2.2 Statistical Validity and the Need to Restrict Data Use. Regardless of the solution, though, this problem of adaptive data reuse makes it clear that data cannot be freely shared and reused without consequence—in a very real sense, every dataset carries with it

[8] Greg Donahue, *They All Got Mysterious Brain Diseases, They're Fighting to Learn Why.* N.Y. Times (Aug. 14, 2024), https://www.nytimes.com/2024/08/14/magazine/canada-brain-disease-dementia.html.

a "statistical validity budget" denoting its capacity to be used for statistically valid scientific inquiry, and uses of the dataset spend from this budget. In this way, data are a resource that is spent as they are used.

It is important to note that adaptivity concerns arise beyond the uses of a single dataset by a single researcher or research lab. As soon as multiple data users are making use of the same or overlapping datasets (perhaps because the data are particular, valuable, expensive, or difficult to collect) and some aspects of their analyses are made visible to the other data users (perhaps via publications, or perhaps via more subtle actions such as strategic business decisions), the concerns stemming from adaptivity arise. Thus, the statistical validity budget associated with a dataset is actually a budget shared (and potentially competed or negotiated over) by multiple parties that wish to use the data. That is, data are a rival good.

3 Data in the Law

In recent years, the landscape of privacy and data protection has undergone a deep transformation, with various countries and jurisdictions enacting comprehensive privacy and data protection frameworks. These regulations focus primarily on the relationship between individuals and their data, providing data subjects with rights over their data, as well as the ability to make decisions regarding various aspects of their data, including regarding their collection, use, and sharing. A central aspect of these laws is the emphasis on user autonomy [41]. By granting individuals rights over their data such as access to their data, the ability to request correction or deletion of their data, or to know and understand the goals for which their data are collected, stored and used, regulators have sought to increase individual control over their data and to ensure that it is individuals, and not large corporations or platforms, that are making decisions regarding user data.[9]

In 2020, the European Commission published its Strategy for Data, describing the EU's data goals for the next five years. The strategy recognizes that "[d]ata is the lifeblood of economic development" and seeks to promote the creation of "a single European data space . . . where personal and non-personal data, including sensitive business data, are secure and businesses also have easy access to an almost infinite amount of high-quality industrial data, boosting growth and creating value." [1] Several regulations have been enacted implementing aspects of the Strategy, including the AI Act [6], the Data Act [5], the Digital Markets Act [2], the Digital Services Act [3] and the Data Governance Act [4]. Each of these regulations focuses on a different aspect of managing data and restructuring the data ecosystem to promote the goals of the Strategy for Data.

The Strategy for Data bemoans the fact that "[c]urrently, a small number of Big Tech firms hold a large part of the world's data." To promote increased data access and use, the EU passed the Data Governance Act (DGA), applicable since September 2023. The Act notes that the full potential of data cannot be realized in the current ecosystem, as data are siloed, held by different individuals and bodies and largely inaccessible to most actors within the data ecosystem. In light of this, the Act seeks to create an infrastructure that will encourage increased sharing and reuse of data by establishing three legal regimes, each resulting in multiple bodies tasked with collecting and sharing data. First, it seeks to encourage and facilitate the reuse of data held by public sector bodies. Second, it creates a framework for altruistic data-sharing by individuals. Finally, it develops the legal infrastructure for data intermediaries, tasked with increasing trust in data-sharing by providing neutral data intermediation services.

The DGA seeks to encourage the safe reuse of protected data held by public sector bodies. Because of the protected nature of the data, the Act stresses that special care should be taken when sharing these data with third parties. In this vein, the Act notes several approaches for data protection such as anonymization and pseudonymization.[10] Numerous public sector bodies will now be subject to the Act, each of which are responsible for managing and now facilitating the sharing and reuse of large amounts of data.

Another pillar of the DGA is data altruism—encouraging individuals to donate their data voluntarily and with no compensation. Recognizing that individuals want to ensure that they are sharing their data with trusted parties that will respect their data and privacy, the DGA allows non-for-profit entities that adhere to certain privacy protections and transparency requirements to register as data altruism organizations recognized in the EU. There is no limit on the number of data altruism organizations, and they are not required to interact or coordinate in any way.

Finally, the Act establishes data intermediaries that act as neutral facilitators for commercial data sharing, connecting between data holders and data users. Since data intermediaries are to play a key role in increasing trust in data sharing, the DGA places several requirements on them. For example, they may provide the technical and legal infrastructure required for data sharing but may not derive value from the data itself (Recital 28). This provision is intended to prevent data intermediaries from replicating the harms stemming from the current structure of the data ecosystem, where a small number of large companies hold the data of many individuals, using it for their own financial benefit.

In summary, in remodelling the European data ecosystem in an effort to increase data sharing and reuse, the DGA has created a reality whereby numerous entities will hold vast amounts of data, with no infrastructure ensuring coordination between them, allowing for accumulated privacy harms and undermining the statistical validity of results based on these data.

3.1 Misconstruing the Nature of Data

Data protection and privacy regulations enacted over the past several years have put a significant focus on creating rights for data subjects (e.g., deletion, rectification, erasure, and access) in relation to each particular entity that holds their data. Much criticism has been has been directed at this "individual-control" approach [32, 44], and the rival aspects of data highlight yet another problem with this individualistic framework: requiring data subjects to manage their data by asserting their rights vis-á-vis each individual data-holder misses the composition of uses by a particular data holder

[9] Many of the protections enshrined in recent privacy and data protection regulation suffer from various challenges; *see* [17].

[10] Anonymization is often left undefined in regulatory text, or defined problematically. *See* [9].

as well as the potential harms created by the fact that multiple bodies may hold, analyze, and share individuals' data and insights from it. Tasking individual data subjects with these "rights" transfers responsibility onto them for the management of a societally valuable resource, without providing individuals with any tools to measure or understand the potential harms, much less to control or limit them. Ensuring that data sharing and reuse are managed in a way that respects privacy and maintains statistical validity requires shifting responsibility for managing the privacy/validity budget associated with data to the data holders.

However, shifting responsibility to data holders will not be enough to overcome this challenge. A data ecosystem consisting of many uncoordinated bodies that hold, manage and share data—as is established by the DGA—reflects a perception of data as a non-rival good; it assumes that the same data can be used by many parties for a variety of purposes without adverse consequences. Acknowledging the rival nature of data makes it clear that, even if each body is operating under the best practices of privacy and statistical validity, a multiplicity of bodies holding, managing, sharing and facilitating the reuse of data, could lead to harmful outcomes. The use of multiple datasets, some of which may include overlapping data, or different categories of data about the same data subjects, could result in an accumulation of privacy harms whereby re-users could cause private data to be leaked or individuals to be re-identified, despite those harms remaining undetectable to the separate data re-users.[11] From a statistical validity perspective, it is hard to interpret the conclusions of an uncoordinated, adaptive analysis of multiple datasets. Unfortunately, many data re-users may not be aware of the potential harmful impact that could arise from their reuse of data. But these challenges exist and cannot be addressed at the level of specific institutions, even when researchers and data re-users are well-intentioned and aware of the potential privacy and statistical validity harms.

4 Reimagining Data Use and Sharing

Recognizing that data are rival in key settings requires us to rethink the way data are collected, held, shared, and governed within the data ecosystem. Any regulation reshaping the data ecosystem should be based on an understanding of the risks to privacy and statistics introduced by unlimited, uncoordinated, or unsupervised sharing and use of data. Data hold enormous potential for societal benefit, but protecting and promoting that benefit requires new policy and technical infrastructure. This section offers principles that should guide the re-imagining of the data ecosystem to facilitate the treatment of data as a rival good. There remain many questions regarding the technical tools that would need to be developed and used to promote the treatment of data as a rival good. While policymakers debate the proper approach, it is imperative to push the research forward.

Even within a single organization or scientific lab, the rival nature of data stemming from privacy and statistical validity concerns raises the need to measure and account for the impact of each computation on privacy/validity and to track the accumulation of such impacts. The toolkits that have been developed in the differential privacy and adaptive data analysis literature provide a starting point, offering algorithms with provably bounded privacy/validity impacts and a mathematical toolkit for tracking their composed effects. However, there is much room for more research in these directions, to develop algorithms for a variety of tasks that provably provide the best possible tradeoffs.

The use of a rival resource across multiple parties highlights the need for systems that allow coordination of data uses across many users, and possibly across multiple datasets. In some settings, a designated trusted party could assume the role of coordination. A distributed approach to such coordination could be supported by new deployments of secure multiparty computation technologies, allowing multiple parties to all be aware of the total accumulated usage of the data without needing to share details of the underlying computations.

In settings where data users cannot be trusted to limit or properly track and report their data usage, it may be more desirable to place data under the curation and protection of data co-operatives, which could be tasked with carrying out the functionalities of limiting and tracking the accumulation of privacy and validity impacts. Such coordination bodies could also serve as clearinghouses that promote (responsible) data accessibility.[33, 34, 38]

Data curators could intervene to reduce unnecessary uses of data, for example, by providing a service that could allow potential data users to check (in a privacy/validity-preserving fashion) whether specific data are indeed a good fit for their use case. Another role for data curators could be to create privacy-preserving data summaries or synthetic datasets, which can be subjected to unlimited analyses across unlimited data users without further privacy degradation; the disadvantage of this approach is that it only guarantees accuracy/validity for a fixed-in-advance set of computations.

An ambitious technical direction is to identify whether it is possible to collect *telemetry* information about data usage, analogously to measurements done in environmental sciences, which would allow monitoring the loss of privacy and statistical validity and inform decisions on which uses to allow and which to limit. This proposal still requires a large amount of research for it to be implementable in practice.

On the policy side, regulation can play a role, promoting or requiring the use of technologies that provide provable or trackable privacy/validity impacts, especially in settings where privacy is at stake or where the data being used were gathered using public funds or otherwise hold particular promise for societal benefit. Data curators who make data available for reuse to multiple parties should also be held responsible for deploying such technologies to protect the privacy and validity of the data they host.

Policy needs to acknowledge that the impacts of data use and the accumulation of harms go beyond the walls and data silos of any one organization, and it is insufficient to provide guarantees that only consider the data uses of a single entity. Regulation could play an important role in establishing coordinating bodies (and coordinating across them when multiple such bodies exist) and endowing them and data holders with rights and responsibilities. In

[11] Article 5(5) of the DGA, prohibiting re-users from re-identifying individuals whose data they received access to is of little help for two main reasons. First, the re-user may not intend or notice that they have uncovered information that, perhaps combined with information released by other data re-users, can re-identify an individual. Second, the re-identification may be done by a third party based on findings made public by re-users.

addition, regulation could determine whose interests a coordinating body should promote and to whom it would owe a duty of care.

A coordinating body or other domain-relevant authorities could and should also be involved in the exercise of setting budgets and striking tradeoffs that determine how much use a dataset should sustain before the risks of further use outweigh the benefits—e.g., when is a bit more privacy loss worth a more accurate computation that will result in more efficient allocation of public resources? Introducing this notion of a budget and tradeoffs and the concept that certain properties of a dataset are "spent" as data are used would be a powerful shift in the regulatory space.

The policy directions outlined above do not currently align with the approach promoted by the DGA, since the DGA promotes unfettered sharing and reuse by data holders and supports the creation of various bodies that would hold and control access to data *independently* of each other. The DGA approach risks undermining the privacy and statistical validity of the data whose sharing and reuse it regulates. Until policymakers adopt a coherent stance on this matter, there is still much research and data collection that can and should be undertaken. The more information and technical tools policymakers have at their disposal when the time comes for them to make a decision, the more flexible they can be in crafting a regulatory response that will facilitate the treatment of data as a rival good.

In many cases, the full set of computations that all parties might wish to carry out on a dataset will be more than the budget allows. The rival characteristics of data reveal the need for infrastructure and technologies that will allow negotiation between various data users and prioritization and selection of computations—e.g., how to negotiate a tradeoff between getting a slightly more accurate count of the number of single-parent households in a region or a slightly more accurate count of the number of families below the poverty line. Here, the technical infrastructure is highly underdeveloped. One possible direction to explore is the development of data markets that would transact on various rights around data related to privacy and statistical validity budgets. The lesson for policy is that there is a need to more precisely articulate the goals and objectives of using data, and to realize that tradeoffs between these various objectives will inevitably arise; mechanisms must be put in place to recognize and negotiate such tradeoffs.

Beyond specific technical and policy interventions, recognition of data's rival aspects calls for a change of culture. Many recent policy initiatives focus on promoting the sharing and accessibility of data, and many recent efforts in the academic space focus on posting and sharing datasets for the advancement of science. When the data are personal, such sharing can be problematic from a privacy perspective. Even when the data do not pertain to humans, though, if they will be used to train machine learning models or to draw conclusions about trends, correlations, or other broader phenomena, then unprotected data sharing carries a risk that it can result in a wave of erroneous scientific "findings" that can pollute the scientific discourse. In place of pushes for broad data posting and sharing, in settings where privacy or statistical validity are of importance, the rival nature of data indicates the importance of prioritizing a culture of stewardship and responsibility.

5 Conclusion

We have argued that data do have rival characteristics in at least two aspects, challenging a broadly held view of data as a non-rival good. In the context of privacy, the ongoing use of data continuously leaks more information about the dataset, depleting its privacy budget. From a statistical perspective, the adaptive reuse of data creates risks for the validity of the conclusions of statistical analyses, endangering the progress of scientific understanding. These concerns should be reflected in regulation restructuring the data ecosystem and, in particular, in any regulation attempting to encourage the reuse of data. We observe that in its current state, the EU Data Governance Act, part of the European Strategy for Data, risks undermining its own goal of ensuring the safe sharing and reuse of data.

Policy based on recognition of the rival nature of data should involve a coordinated effort to protect and reuse data, supervised, and managed by a designated body (or bodies). Technical and legal tools should be put in place to ensure the successful operation of this body and its integration into the data ecosystem.

As generative AI continues to develop, increasingly demanding new fresh sources of data, we are likely to face even greater challenges. Restructuring the data ecosystem based on a deep understanding of the nature of data has the potential to transform the way data are managed and help society navigate a future in which data scarcity is the new reality.

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